

Within-Firm Variation and Economic Magnitude

TEACHING NOTE

ACCTG 597A • Week 7

On Mummolo and Peterson (2018), “Improving the Interpretation of Fixed Effects Regression Results,” *Political Science Research and Methods* 6(4): 829–835. Running example: Anantharaman and Lee (2014, *JFE*).

A&L report magnitudes per pooled SD

- A&L scale CFO vega by its pooled SD, in their firm-effects tables too.

“In tests, we standardize all compensation measures by subtracting the sample mean and scaling by the sample standard deviation, allowing easy assessment and comparisons of economic significance.”

Anantharaman and Lee (2014, p. 334)

Bet for the room: the share of the pooled SD left after fixed effects in published studies.

1.76 points

underfunding per pooled SD of CFO vega: A&L Table 4A, Model 3, at the 95th percentile of distress

15.8%

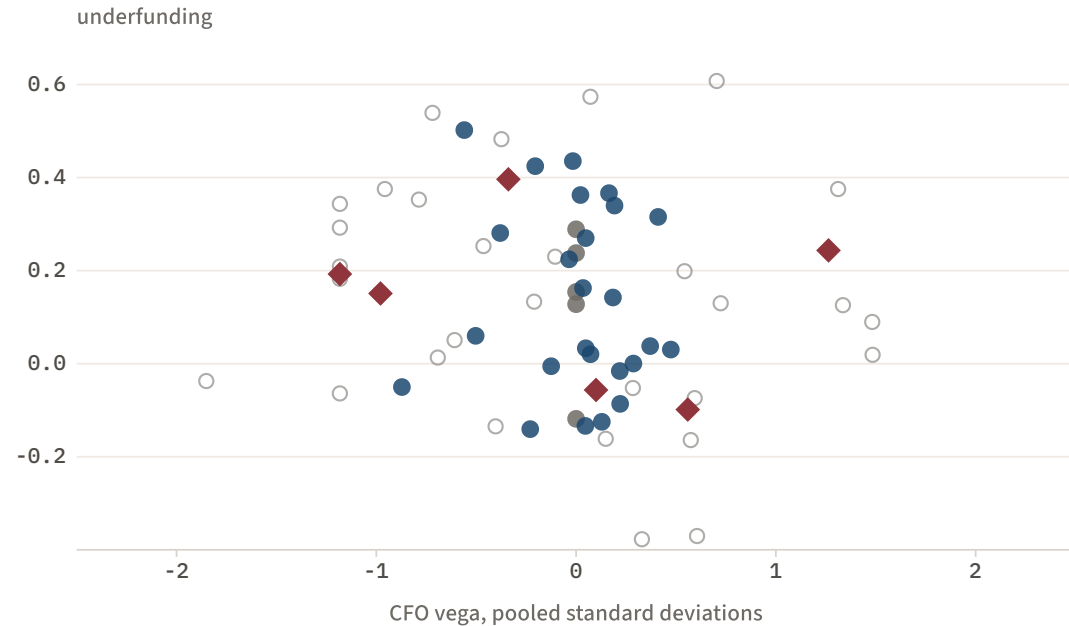
mean underfunding (A&L Table 1)

Firm effects remove between-firm variation

- Subtracting each firm's mean removes the spread across firms and leaves only movements within each firm.

Maroon diamonds mark firm means; the navy deviations around them are all the estimator sees.

Simulated panel (seed 2014): 900 firms $\times$ 5 years; 25 percent of vega's variation within firms, assumed.



☒ Subtract each firm's mean

Navy: deviations from each firm's mean, SD 0.50 pooled SDs. Maroon: the firm means the estimator discards

Six firms drawn from the simulated panel (seed 2014). Grey firm: CFO vega never changes. Hollow circles: pooled positions. After the toggle both axes show deviations from firm means, re-centred at the sample mean.

The estimator sees only within deviations

M&P Eq. (1)

$$Y_{it} = \alpha_i + \beta X_{it} + \varepsilon_{it}$$

M&P Eq. (2): subtract each firm's mean

$$Y_{it} - \bar{Y}_i = \beta (X_{it} - \bar{X}_i) + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

OLS on Eq. (2), with $\tilde{x}_{it} = X_{it} - \bar{X}_i$

$$\hat{\beta} = \frac{\sum_i \sum_t \tilde{x}_{it} (Y_{it} - \bar{Y}_i)}{\sum_i \sum_t \tilde{x}_{it}^2}$$

Sum of squares of X; T_i = years for firm i

$$\underbrace{\sum_{i,t} (X_{it} - \bar{X})^2}_{\text{total}} = \underbrace{\sum_{i,t} \tilde{x}_{it}^2}_{\text{within}} + \underbrace{\sum_i T_i (\bar{X}_i - \bar{X})^2}_{\text{between}}$$

- The fixed effects slope uses only deviations from firm means, whose variance never exceeds the total.

“Equation 2 makes plain that β is estimated solely based on deviations from each unit’s average treatment value over time.”

Mummolo and Peterson (2018, p. 830, fn. 3)

A calibrated panel reproduces the algebra

- Three routes give one slope; with 25 percent of the variation within firms, the within SD is half the pooled.

```
1 xtreg uf vega i.year, fe vce(cluster firm)
2 regress vega i.firm i.year
3 predict double vt, residuals // x~: vega net of firm and year means
4 regress uf vt // same slope (Frisch-Waugh-Lovell, Week 3)
5 summarize vega vt // SD(x) and SD(x~)
```

Year effects remove still more: in M&P’s words, “the variance reduction is even more severe” (p. 830).

Simulated panel (seed 2014)	Value
SD(x), pooled	1.000
SD($\tilde{x}$), firm effects: $X - \bar{X}_i$	0.500
SD($\tilde{\tilde{x}}$), firm and year effects: $X - \bar{X}_i - \bar{X}_t + \bar{X}$	0.466
Slope, firm and year effects	0.0158 (t = 2.47)
Three routes, largest difference	0.0000000000
SST – SSW – SSB	0.0000000000

mp_within_variation.do: underfunding = firm trait + 0.0176 × vega + noise; 0.0176 is A&L’s slope at the 95th percentile of distress.

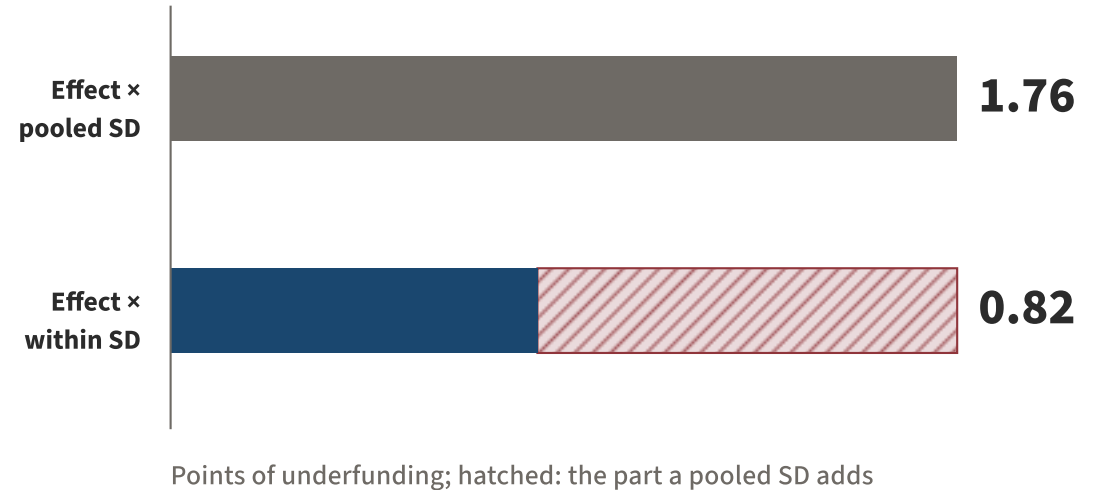
A within SD shrinks the described effect

- A one-within-SD change implies an effect smaller by a factor equal to the square root of the share left.

“Multiply the estimated coefficient of interest by the revised standard deviation of the independent variable to assess substantive importance.”

Mummolo and Peterson (2018, p. 834)

At the simulated share left (21.7 percent), 1.76 points becomes **0.82** points per within SD.



Share left



21.7%

Within SD / pooled SD = $\sqrt{\text{share}} = 0.47$. Simulated panel: 21.7% left after firm and year effects; M&P's five studies: 6% to 67%

Effect per pooled SD: 1.76 points, the marginal effect of CFO vega at the 95th percentile of distress, computed from A&L Table 4A, Model 3 ($22.0 + 2.340 \times -1.877$, per 1,000). The share left is not reported by A&L; M&P range from their Table 1, squared ratios.

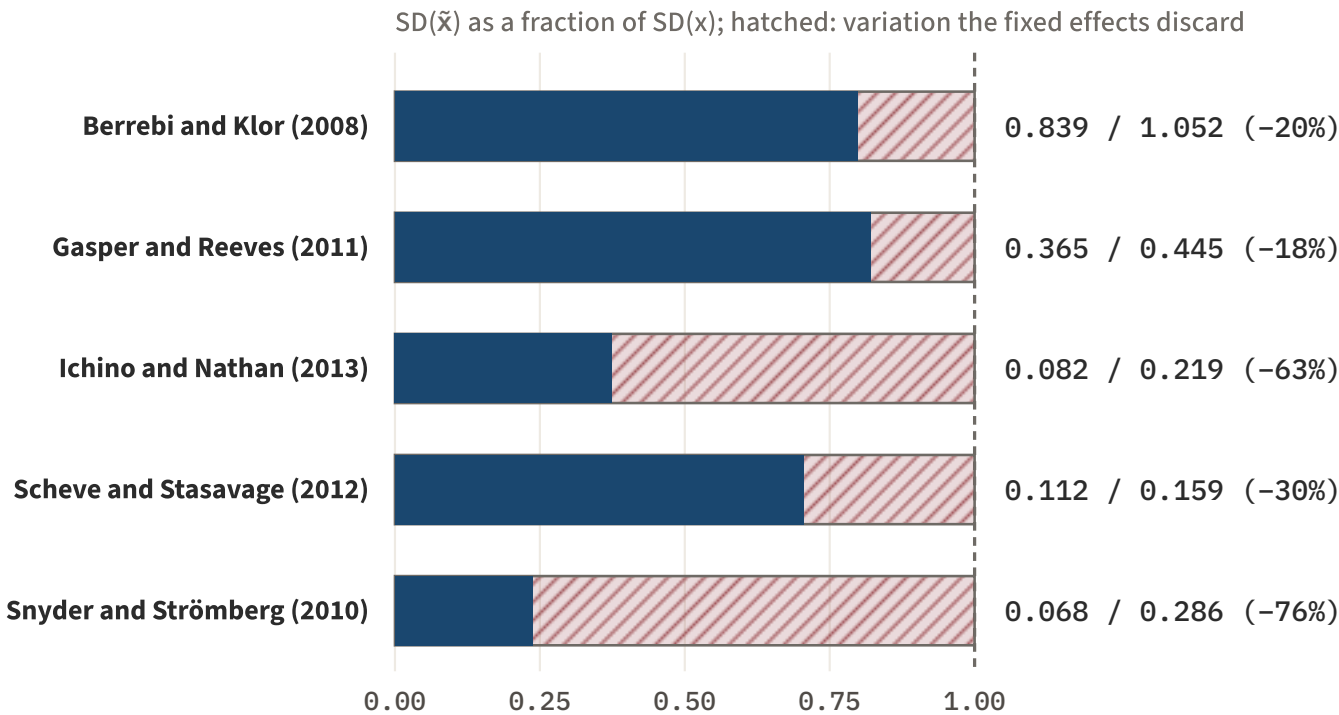
One-SD effects shrink by 18 to 76 percent

- Some studies keep most of the pooled variation and others under a quarter; the estimator does not predict which.

In Snyder and Strömberg (2010), “the standard deviation of the variable drops from 0.286 to 0.068, a reduction of 76 percent” (M&P p. 831).

A&L do not report the corresponding figure for CFO vega.

Room: [stated shifts and the within SD](#)



Navy: SD after residualizing on each study's fixed effects, as a fraction of the pooled SD (grey outline = 1). Maroon hatching: the share of the pooled SD the fixed effects discard. Mummolo and Peterson (2018), Table 1 (p. 832), columns 2 to 4.

Firm effects give constant firms no weight

Firm effects only (K&K Eqs. 10–11); $\hat{\beta}_i$ is firm i 's own slope

$$\hat{\beta} = \frac{\sum_i w_i \hat{\beta}_i}{\sum_i w_i}, \quad w_i = \sum_t (X_{it} - \bar{X}_i)^2$$

- Under firm effects alone, a firm's weight is its within sum of squares; a constant-vega firm gets none.

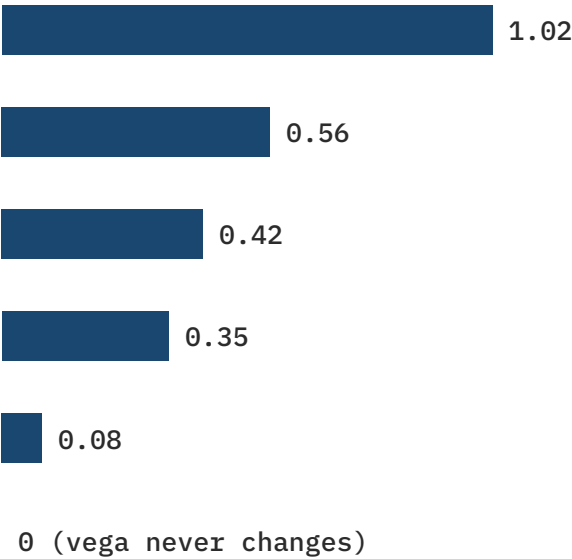
M&P: “units without variation provide no information during one-way unit-fixed effects estimation” (p. 829). With year effects, a constant firm stays in as a comparison, like a never-treated firm in a DiD.

Simulation: 72 of 900 firms constant (assumed); with year effects they hold 1.2% of the two-way sum of squares. Room: [deviation versus change](#).

Five years of CFO vega (pooled SDs)



Weight: within sum of squares



Six firms from the simulated panel; hollow maroon diamonds are firm means. Weights under firm effects only. In the full panel 72 of 900 firms have constant vega (an assumption).

M&P’s checklist changes reporting only

■ Each item adds a reported quantity; the coefficient and its t-statistic are unchanged.

M&P checklist (pp. 833–834)	A&L, as published	Illustration: simulated panel, assumed share
1. Residualize vega on the fixed effects	not reported	SD($\tilde{x}$) = 0.47 pooled SDs
2a. Effect $\times$ SD($\tilde{x}$); share without variation	coefficient per pooled SD; no magnitude computed from Table 4	0.82 points per SD($\tilde{x}$); 8% of firms constant
2b. A stated shift relative to SD($\tilde{x}$)	50th to 95th percentile = 1.99 pooled SDs, applied to Table 3 (pp. 335–337)	4.27 SD($\tilde{x}$)
3. Describe the variation in within terms	A&L Table 6 (pension equity): “within firms over time” (p. 340)	the same phrasing for Table 4

“Our critiques also have no bearing on the statistical significance of any published results.”

Mummolo and Peterson (2018, p. 830, fn. 2)

Room: the outcome scale.

The variation left defines the magnitude

- 1** A fixed effects coefficient uses only the variation the fixed effects leave; its magnitude should be stated per standard deviation of that variation, often well below the pooled one.
- 2** Under firm effects, firms whose regressor never changes contribute nothing; with year effects they serve only as comparisons.
- 3** The coefficient and its t-statistic are unchanged; the magnitude and the firms it describes depend on the variation left.

Mummolo and Peterson (2018) · Anantharaman and Lee (2014) · github.com/sbb151/acctg597a-research-design

Appendix

Stated shifts reach fifteen within-unit SDs

- Any counterfactual should be reported as a multiple of the within SD.

M&P Table 1, column 5, divides the shift each study discussed by $SD(\tilde{x})$: 1.192, 2.736, 2.667, 8.953, and 14.768 (p. 832); shifts of the last two sizes “are rarely observed in the data” (p. 833).

A&L’s 50th-to-95th percentile shift in CFO vega is $(202.77 - 19.87)/91.77 = 1.99$ pooled SDs (Table 1; pp. 335–337), or 4.27 $SD(\tilde{x})$ at the simulated share.

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1.2 to 14.8

stated shift / $SD(\tilde{x})$, M&P Table 1, column 5

4.27

A&L’s 50th-to-95th shift in $SD(\tilde{x})$ units (simulated share)

A deviation from a mean is not a change

- With independent within-firm news, $SD(\tilde{x})$ is smaller than a typical change between two years.

With independent news of variance σ^2 , the deviation from a five-year mean has standard deviation $\sigma\sqrt{4/5} \approx 0.89\sigma$, whereas a year-to-year change has $\sigma\sqrt{2} \approx 1.41\sigma$; with persistent within-firm movements the comparison can differ. M&P call $SD(\tilde{x})$ “a typical shift” (p. 834); their checklist also asks for a histogram of within-unit ranges (item 2), a direct description.

Constant-vega firms enter $SD(\tilde{x})$ as zeros under firm effects, so the pooled $SD(\tilde{x})$ also understates the typical deviation among firms that change.

A critique developed for this note, not a claim made by M&P. [Back](#)

The outcome scale is a reporting choice too

- M&P's fourth item asks for effects in outcome units and in SDs of the original and residualized outcome.

“Consider characterizing this new effect in terms of both the outcome’s units and in terms of standard deviations of the original and transformed outcome (i.e., the outcome residualized with respect to fixed effects).”

Mummolo and Peterson (2018, p. 834)

M&P leave the choice to the substance of the study (p. 834). This note adds that fixed effects also shrink the outcome’s variance, so effects stated in residualized-outcome SDs grow mechanically.

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